

Breathing In, Breathing Out — Paper C

HARD • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided.
- A calculator is allowed, but every number here is designed to work without one.
- Where a question asks about air moving in or out, your answer must follow the chain in order: muscles change the volume, the volume change causes a pressure change, air then flows down the pressure gradient. A muscle never sucks or pushes air by itself.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe experiments and data you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- Never write that respiration takes place in the lungs. Respiration is a reaction inside cells; the lungs carry out ventilation and gas exchange.

SECTION 1 – THE SIZE OF THE SURFACE

- 1.** An adult lung holds about 3.0×10^8 alveoli. Each alveolus has a surface area of about $1.6 \times 10^{-7} \text{ m}^2$. A single smooth sac holding the same volume of air would have a surface area of 0.06 m^2 .

- (a)** Calculate the total surface area of the alveoli, in m^2 . [2]

- (b)** Calculate how many times larger this is than the single smooth sac. [1]

- (c)** Explain why dividing the same volume of lung into millions of tiny sacs speeds up gas exchange. [2]

Multiply the numbers in front first, then add the powers of ten.

- 2.** Oxygen crosses about 1 micrometre of tissue on its way from alveolar air into the blood.

- (a)** One micrometre is one millionth of a metre. Write this distance in metres, in standard form. [1]

- (b)** A disease thickens this barrier so that oxygen must cross twice the distance. Explain the effect on the person. [2]

SECTION 2 – READING SOMETHING YOU HAVE NOT SEEN BEFORE

- 3.** A spirometer records the volume of air a person breathes as a moving trace on a chart. A trace taken while a patient sat still showed 6 complete breaths in 30 seconds, and each breath moved 0.45 dm^3 of air.

- (a)** State the volume of air moved in one breath. This is called the tidal volume. [1]

- (b)** Calculate the patient's breathing rate, in breaths per minute. [1]

- (c)** Calculate the total volume of air breathed in one minute, in dm^3 . [2]

- (d)** The patient then walks on a treadmill. Describe the two ways the trace would change, and say what each one shows. [2]

- 4.** A model of the thorax is built from a sealed glass bell jar with a rubber sheet stretched across its open base, and two balloons hanging inside from a Y-shaped tube through the lid. Pulling the rubber sheet downwards makes the balloons inflate.

- (a)** State what the rubber sheet, the glass jar and the balloons each represent in a real person. [3]

- (b)** Give one way in which the model is unlike a real thorax. [1]

SECTION 3 – WHEN THE OXYGEN RUNS SHORT

- 5.** A sprinter runs 200 m flat out.

- (a)** Write the word equation for the respiration taking place in her leg muscles near the end of the race. [1]

- (b)** Give one advantage and one disadvantage of this type of respiration compared with aerobic respiration. [2]

- 6.** She stands bent over, breathing hard, for almost a minute after she has stopped running. Explain why, using the term oxygen debt. [3]

- 7.** Two students measure their breathing rates, in breaths per minute, before exercise, at the end of exercise, and three minutes after stopping.

	Before	At the end	After 3 min
Student P	16	40	17
Student Q	20	45	30

- (a)** Calculate the percentage increase in Student P's breathing rate during exercise. [2]

- (b)** State which student is fitter, and give two reasons from the data. [2]

- (c)** Both students are still above their resting rate at the end of exercise. Explain why breathing does not return to normal the moment exercise stops. [2]

Percentage increase is measured against the value you started from, not the value you finished on.

End of paper. Check every answer has a unit or a form the question actually asked for.